

BAYANTOOROI, MongoliaWMO: **443250**Lat: **44.924N**Lon: **96.753E**Elev: **2181**StdP: **77.72**Time Zone: **7.00 (E07)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.3	-29.7	-35.6	0.2	-30.8	-32.8	0.2	-26.5	15.6	-13.3	10.3	-11.4	1.3	90	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	34.8	18.2	33.3	16.9	31.8	16.2	21.2	30.7	19.8	28.7	18.6	27.5	3.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	17.7	26.0	17.2	16.1	24.4	15.7	14.6	24.4	73.6	31.1	67.7	28.6	62.9	27.6	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	11.9	10.1	DB	-31.4	37.2	4.9	1.3	-35.0	38.1	-37.8	38.9	-40.6	39.6	-44.2	40.5	(4)
(5)			WB	-32.0	21.5	4.8	2.7	-35.4	23.4	-38.2	25.0	-40.9	26.6	-44.3	28.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-17.6	-11.9	0.5	10.2	16.2	22.8	25.2	23.3	16.2	7.0	-4.3	-14.1	(6)	
	DBStd	15.51	6.27	7.05	6.27	5.11	4.17	3.35	3.05	3.14	3.90	4.40	5.55	5.50	(7)	
	HDD10.0	3118	855	612	295	58	5	0	0	0	4	113	428	748	(8)	
	HDD18.3	4974	1113	845	552	244	90	6	1	2	84	353	678	1007	(9)	
	CDD10.0	1739	0	0	1	65	197	384	471	411	191	19	0	0	(10)	
	CDD18.3	554	0	0	0	2	23	140	213	155	21	0	0	0	(11)	
	CDH23.3	6321	0	0	0	25	306	1591	2502	1672	225	1	0	0	(12)	
	CDH26.7	2623	0	0	0	3	74	643	1179	682	42	0	0	0	(13)	
(14) Wind	WSAvg	3.1	1.8	2.3	3.5	4.4	4.6	3.8	3.6	3.5	3.0	2.8	2.3	2.0	(14)	
(15) Precipitation	PrecAvg	48	1	1	1	2	1	6	17	12	4	1	1	1	(15)	
	PrecMax	138	8	6	4	10	7	26	57	57	20	11	6	4	(16)	
	PrecMin	9	0	0	0	0	0	0	1	0	0	0	0	0	(17)	
	PrecStd	28	2	1	1	3	2	7	15	14	6	2	1	1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.4	7.2	17.5	26.4	31.5	35.5	37.4	35.5	30.1	21.7	11.9	0.5	(19)	
		MCWB	-5.4	0.5	5.8	11.9	13.4	18.0	19.2	17.8	13.2	11.0	4.2	-4.0	(20)	
	2%	DB	-4.7	3.3	15.2	23.4	29.0	33.6	35.6	33.5	27.6	19.1	8.8	-2.5	(21)	
		MCWB	-7.7	-2.2	4.7	10.1	13.9	17.1	19.1	16.2	12.7	8.8	2.7	-5.8	(22)	
	5%	DB	-7.0	0.5	13.1	21.6	26.9	32.0	33.9	32.1	25.9	17.1	6.1	-4.5	(23)	
		MCWB	-9.5	-4.0	3.7	9.3	12.9	16.1	18.1	16.0	12.5	7.8	1.0	-7.4	(24)	
	10%	DB	-9.4	-1.8	10.7	19.5	24.8	30.3	32.4	30.5	24.0	15.1	3.7	-6.7	(25)	
		MCWB	-11.4	-5.4	2.4	8.2	12.3	15.8	16.9	15.8	11.7	6.8	-0.7	-9.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.2	0.5	6.9	13.3	17.5	21.3	24.0	21.6	16.4	12.7	4.9	-3.7	(27)	
		MCDB	-1.7	6.7	15.5	23.4	26.2	30.9	33.4	29.2	23.6	19.5	10.3	-0.1	(28)	
	2%	WB	-7.6	-2.1	5.3	11.5	15.6	19.8	22.2	20.0	15.0	10.5	3.0	-5.6	(29)	
		MCDB	-4.7	2.9	14.0	21.6	26.0	29.3	32.2	28.4	22.6	17.6	8.5	-2.7	(30)	
	5%	WB	-9.5	-3.7	4.0	10.1	14.3	18.7	20.7	18.8	13.9	8.7	1.2	-7.3	(31)	
		MCDB	-7.1	0.0	12.7	20.0	24.5	28.4	30.6	27.0	22.9	15.5	6.0	-4.6	(32)	
	10%	WB	-11.3	-5.4	2.6	8.8	13.0	17.6	19.2	17.6	12.9	7.1	-0.7	-9.0	(33)	
		MCDB	-9.4	-1.8	10.5	18.5	23.2	26.9	27.9	26.8	21.6	14.2	3.5	-6.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	13.4	14.7	13.8	13.1	12.5	12.0	12.1	13.1	13.8	12.3	11.1	(35)	
		MCDBR	13.5	15.1	15.8	15.5	15.7	14.5	13.4	13.3	14.4	15.5	14.6	12.6	(36)	
		MCWBR	11.1	11.0	9.2	8.4	7.7	6.4	5.4	5.4	6.5	8.7	10.0	10.0	(37)	
	5% WB	MCDBR	13.4	14.7	14.9	14.8	13.6	12.6	11.8	11.2	13.1	14.1	14.4	12.6	(38)	
		MCWBR	11.0	10.8	8.9	8.9	8.2	6.5	6.2	5.8	7.0	8.9	10.1	10.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.232	0.260	0.295	0.330	0.339	0.355	0.351	0.330	0.300	0.274	0.247	0.232	(40)	
	taud		2.435	2.371	2.287	2.246	2.280	2.308	2.356	2.403	2.450	2.456	2.456	2.428	(41)	
	Ebn at Noon		917	939	940	930	928	911	910	920	926	910	886	879	(42)	
	Edh at Noon		69	89	113	129	129	126	119	109	95	81	67	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.04	3.15	4.74	6.19	7.23	7.14	6.88	6.20	5.29	3.78	2.35	1.77	(44)	
	RadStd		0.19	0.20	0.29	0.29	0.39	0.41	0.29	0.32	0.19	0.15	0.18	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	-3.28	N/A	-2.26	-2.41	N/A	N/A	N/A	N/A

Nomenclature: See separate page